THE ASSOCIATION OF OBESITY ON METABOLIC DISEASES AMONG D.C. INPATIENT POPULATION – (2017 – 2019)*

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Introductions

- The obesity epidemic globally impacts morbidity and mortality (James et al., 2001) (Jiang et al., 2016).
- With numerous comorbidities adversely impacting the chronic disease epidemic (Clark & Brancati, 2000).
- Some of these comorbidities are metabolic diseases.

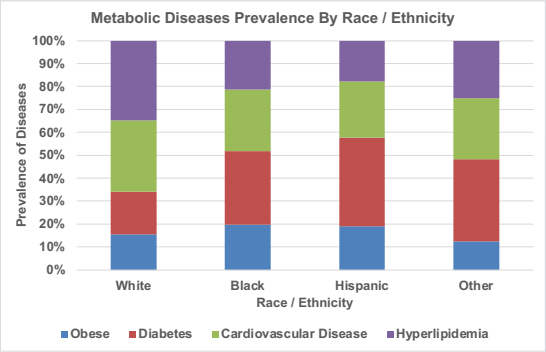
Objective/Purpose

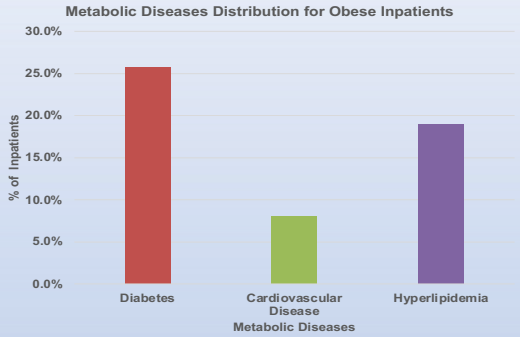
- Our aims were 1. to identify the distributions of metabolic diseases among obese inpatients by race, by payors, and finally by DCI (Distressed Community Index) quantiles, respectively; 2. to determine the impact of obesity as a predictor of the selected metabolic diseases with race, age, and payors as confounders amongst resident inpatients in DC.
- Insights from our study could inform practitioners of the importance of obesity in its role in metabolic disease complications.

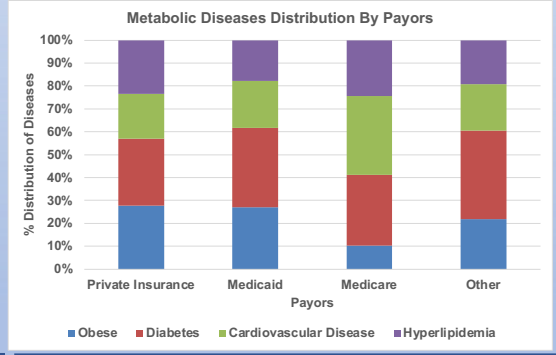
Materials/Methods

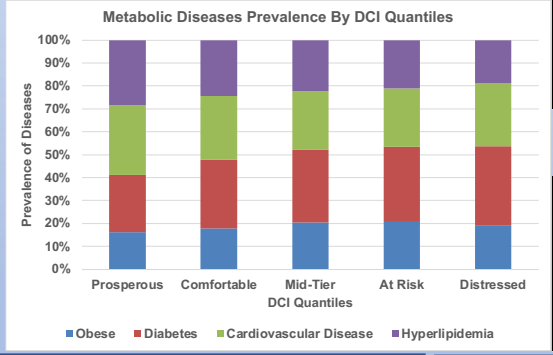
- Our primary data source was the Healthcare Cost and Utilization Project (HCUP) for DC from 2017 to 2019, with inpatients ages 18 to 95, excluding pregnant women and including primary residents.
- The data was cleaned and prepared for analysis using data science techniques with the Python program.
- The disease distributions were calculated using cross-tabulations.
- Logistic regression models to examine the effect of obesity as a predictor for metabolic diseases were also performed.

Results









Results of Logistic Regression Models

The Data was Split by Gender and Race

(Tables are summarized with results of the main predictor variable "OBESITY" only)

All Models were Controlled For:

Payors, Distressed Community Index (DCI) Quantiles, Obesity * Payors, Obesity * DCI Quantiles

Obesity as a Predictor for Cardiovascular Diseases

		Male				Female			
Race / Ethnicity	Main Predictor	Odds Ratio (OR)	P-Value < 0.05	95% C.I.for OR Lower	95% C.I.for OR Upper	Odds Ratio (OR)	P-Value < 0.05	95% C.I.for OR Lower	95% C.I.for OR Upper
White	Obesity	0.658	0.238	0.328	1.318	2.073	0.173	0.727	5.911
Black	Obesity	0.639	0.103	0.373	1.094	0.610	0.018	0.404	0.920
Hispanic	Obesity	1.496	0.467	0.505	4.427	0.738	0.714	0.145	3.745
Other	Obesity	0.474	0.518	0.050	4.542	0.000	0.997	0.000	

Obesity as a Predictor for Diabetes

		Male				Female			
Race / Ethnicity	Main Predictor	Odds Ratio (OR)	P-Value < 0.05	95% C.I.for OR Lower	95% C.I.for OR Upper	Odds Ratio (OR)	P-Value < 0.05	95% C.I.for OR Lower	95% C.I.for OR Upper
White	Obesity	2.336	0.004	1.310	4.166	9.564	0.000	5.248	17.432
Black	Obesity	1.962	0.000	1.368	2.814	2.291	0.000	1.773	2.960
Hispanic	Obesity	1.657	0.263	0.685	4.010	2.054	0.069	0.946	4.456
Other	Obesity	1.135	0.872	0.243	5.296	7.163	0.001	2.274	22.564

Obesity as a Predictor for Hyperlipidemia

		Male				Female			
Race / Ethnicity	Main Predictor	Odds Ratio (OR)	P-Value < 0.05	95% C.I.for OR Lower	95% C.I.for OR Upper	Odds Ratio (OR)	P-Value < 0.05	95% C.I.for OR Lower	95% C.I.for OR Upper
White	Obesity	1.763	0.015	1.118	2.779	3.647	0.000	2.108	6.312
Black	Obesity	1.976	0.001	1.328	2.942	2.089	0.000	1.550	2.816
Hispanic	Obesity	3.761	0.004	1.532	9.233	1.996	0.153	0.773	5.156
Other	Obesity	3.180	0.083	0.860	11.758	2.553	0.260	0.499	13.062

Grant Support

This work is supported by Howard University Research Centers for Minority Institutions (HU-RCMI) with a research grant from the National Institute of Health – National Institute on Minority Health and Health Disparities (NIH-NIMHD) (U54MD007597).

* Each record was counted as an inpatient.

Distressed Community Index Notes

The Distressed Community Index (DCI) was derived by the Economic Innovation Group (EIG). It is a bipartisan public policy organization forging a more dynamic and inclusive American economy. EIG produces nationally recognized research and works with policymakers to develop ideas that empower workers, entrepreneurs, and communities. DCI highlights geographic inequality in the United States at the zip code level. It combines seven complementary economic indicators into a single score: No High School Diploma (Percent of the 25-year-old+ population without a high school diploma or equivalent), Housing Vacancy Rate (Percent of habitable housing that is unoccupied, excluding properties that are for seasonal recreational or occasional use), Adults not working (Percent of the prime age (25 – 54) population not currently employed), Poverty Rate (Percent of the population living under the poverty line), Median Income Ratio (Median household income as a percent of metro area median household income or state for non-metro areas), Change in employment (Percent change in the number of jobs from 2016 – 2020), Change in Establishments (Percent change in the number of business establishments from 2016 – 2020). It is derived from the U.S. Census Bureau's Business Patterns and American Community Survey 5-Year Estimates for 2016-2020. It sorts zip codes into quintiles of well-being: prosperous, comfortable, mid-tier, at risk, and distressed. At the top of the economic ladder are America's prosperous zip codes, boasting robust employment and business growth alongside high household earnings and educational attainment. Distressed communities, meanwhile, represent an alternate, left-behind America that bears signs of profound disconnection from the country's overall economic success. These communities span the geographic spectrum and can be found in urban centers, rural heartlands, and suburbia.

Conclusion & References

Conclusion:

- In conclusion, Obesity is a significant influencer of metabolic syndrome.
- Obesity is more closely associated with diabetes and hyperlipidemia than observed for cardiovascular diseases.
- Individuals from racial/ethnic groups display disparities in metabolic syndrome prevalence.

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